DATABASE MANAGEMENT SYSTEM

Sample Data Insertion

Realistic records for all ten tables of the online_shop database

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Course	BCA Game Development
Subject	Database Management System
Assignment	Assignment 7 — Sample Data Insertion
Project	Online Shopping System

Assignment 7 — requirements

- Insert realistic sample data into all the tables created for your proposed system.
- Include sufficient records to simulate a real-world application relevant to your chosen topic.
- Ensure that the inserted data follows the defined constraints and relationships.

01 TASK 1

Approach and Insertion Order

Parent tables are filled before child tables, so every foreign key finds the row it points to.

The data describes an Indian online store: customers from eight cities, a catalogue of 20 products in six categories, and fifteen orders placed between August and October 2026. Prices are in rupees. Every row satisfies the primary key, foreign key, UNIQUE, CHECK and DEFAULT rules defined in Assignments 5 and 6.

Step	Table	Rows	Must be filled after
1	customers	12	—
2	pincodes	10	—
3	addresses	14	customers, pincodes
4	categories	6	—
5	products	20	categories
6	orders	15	customers, addresses
7	order_items	30	orders, products
8	payments	16	orders
9	shipments	12	orders
10	reviews	15	customers, products

Real situations that the data deliberately covers:

Situation	Where it appears in the data
Customer with more than one saved address	Customers 1 and 2 each have a Home and an Office address
Customer who never ordered	Customer 12 (Lakshmi Venkatesh) has no orders and no phone number
Product that has never been sold	Product 20 (Large Mouse Pad)
Product that is out of stock	Product 10 (Introduction to Game Design), stock_qty = 0
All four order statuses	Completed (9), Placed (3), Confirmed (2), Cancelled (1)
Price captured at purchase time	Order 2 bought the monitor at ₹15,499.00; today it costs ₹14,999.00
Failed payment followed by a successful retry	Order 14: payment 14 (Card, Failed), payment 15 (UPI, Paid)
Cancelled order with a refund	Order 8: payment 8 has status Refunded
Order split across two shipments	Order 6: shipments 6 and 7 with different carriers
Paid but not yet shipped / unpaid	Order 14 (paid, no shipment yet); orders 13 and 15 (payment pending)

02 TASK 2

INSERT Statements

One multi-row INSERT per table. Keys are written explicitly so related rows line up.

2.1 customers

```
INSERT INTO customers (customer_id, name, email, phone) VALUES
  (1, 'Aarav Sharma', 'aarav.sharma@example.com', '9876500001'),
  (2, 'Priya Nair', 'priya.nair@example.com', '9876500002'),
  (3, 'Rohan Mehta', 'rohan.mehta@example.com', '9876500003'),
  (4, 'Ananya Iyer', 'ananya.iyer@example.com', '9876500004'),
  (5, 'Karthik Reddy', 'karthik.reddy@example.com', '9876500005'),
  (6, 'Sneha Patil', 'sneha.patil@example.com', '9876500006'),
  (7, 'Vikram Singh', 'vikram.singh@example.com', '9876500007'),
  (8, 'Divya Menon', 'divya.menon@example.com', '9876500008'),
  (9, 'Arjun Das', 'arjun.das@example.com', '9876500009'),
  (10, 'Meera Kulkarni', 'meera.kulkarni@example.com', '9876500010'),
  (11, 'Imran Khan', 'imran.khan@example.com', '9876500011'),
  (12, 'Lakshmi Venkatesh', 'lakshmi.venkatesh@example.com', NULL);
```

2.2 pincodes

```
INSERT INTO pincodes (pincode, city, state) VALUES
  ('560001', 'Bengaluru', 'Karnataka'),
  ('560034', 'Bengaluru', 'Karnataka'),
  ('560066', 'Bengaluru', 'Karnataka'),
  ('400001', 'Mumbai', 'Maharashtra'),
  ('110001', 'New Delhi', 'Delhi'),
  ('600001', 'Chennai', 'Tamil Nadu'),
  ('500001', 'Hyderabad', 'Telangana'),
  ('411001', 'Pune', 'Maharashtra'),
  ('700001', 'Kolkata', 'West Bengal'),
  ('380001', 'Ahmedabad', 'Gujarat');
```

2.3 addresses

```
INSERT INTO addresses (address_id, customer_id, label, house_street, pincode) VALUES
  (1, 1, 'Home', '12, 4th Cross, Indiranagar', '560001'),
  (2, 2, 'Home', '45, Residency Road', '560034'),
  (3, 2, 'Office', '7, Tech Park Main Road, Whitefield', '560066'),
  (4, 3, 'Home', '203, Marine Lines Apartments', '400001'),
  (5, 4, 'Home', '88, Gandhi Nagar', '600001'),
  (6, 5, 'Home', '14, Jubilee Hills Road No. 5', '500001'),
  (7, 6, 'Home', '31, Model Colony', '411001'),
  (8, 7, 'Home', '9, Connaught Circus', '110001'),
  (9, 8, 'Home', '56, Salt Lake Sector 2', '700001'),
  (10, 9, 'Home', '22, Navrangpura Society', '380001'),
  (11, 10, 'Home', '5, Koregaon Park Lane', '411001'),
  (12, 11, 'Home', '101, Park Street', '700001'),
  (13, 12, 'Home', '18, Brigade Road', '560001'),
  (14, 1, 'Office', '3, MG Road Business Centre', '560001');
```

2.4 categories

```
INSERT INTO categories (category_id, category_name, description) VALUES
  (1, 'Electronics', 'Phones, audio devices and accessories'),
  (2, 'Clothing', 'Men''s and women''s apparel'),
  (3, 'Books', 'Academic and general reading'),
  (4, 'Home & Kitchen', 'Appliances and cookware'),
  (5, 'Sports & Fitness', 'Exercise equipment and activewear'),
  (6, 'Gaming', 'Controllers, keyboards and gaming accessories');
```

2.5 products

```
INSERT INTO products (product_id, category_id, name, price, stock_qty) VALUES
  (1, 1, 'Wireless Bluetooth Earbuds', 1499.00, 120),
  (2, 1, '20000 mAh Power Bank', 1899.00, 85),
  (3, 1, 'Smart Fitness Band', 2499.00, 60),
  (4, 1, '27-inch IPS Monitor', 14999.00, 25),
  (5, 2, 'Cotton Round-Neck T-Shirt', 499.00, 300),
  (6, 2, 'Slim Fit Denim Jeans', 1299.00, 150),
  (7, 2, 'Hooded Sweatshirt', 1799.00, 90),
  (8, 3, 'SQL Fundamentals for Beginners', 599.00, 200),
  (9, 3, 'Database System Concepts', 1250.00, 75),
  (10, 3, 'Introduction to Game Design', 850.00, 0),
  (11, 4, 'Non-Stick Cookware Set', 2299.00, 55),
  (12, 4, 'Electric Kettle 1.5 L', 899.00, 110),
  (13, 4, 'Mixer Grinder 750 W', 3499.00, 35),
  (14, 5, 'Yoga Mat 6 mm', 699.00, 140),
  (15, 5, 'Adjustable Dumbbell Pair', 3299.00, 30),
  (16, 5, 'Running Shoes', 2799.00, 70),
  (17, 6, 'Wireless Game Controller', 3299.00, 45),
  (18, 6, 'Mechanical Gaming Keyboard', 3999.00, 38),
  (19, 6, 'Gaming Mouse', 1599.00, 80),
  (20, 6, 'Large Mouse Pad', 399.00, 60);
```

2.6 orders

```
INSERT INTO orders (order_id, customer_id, address_id, order_date, status, total_amount) VALUES
  (1, 1, 1, '2026-08-03 10:15:00', 'Completed', 2098.00),
  (2, 2, 2, '2026-08-07 14:40:00', 'Completed', 17098.00),
  (3, 3, 4, '2026-08-12 09:05:00', 'Completed', 2796.00),
  (4, 4, 5, '2026-08-19 18:22:00', 'Completed', 3198.00),
  (5, 5, 6, '2026-08-25 21:10:00', 'Completed', 7298.00),
  (6, 1, 14, '2026-09-02 11:30:00', 'Completed', 2598.00),
  (7, 6, 7, '2026-09-06 16:45:00', 'Completed', 2448.00),
  (8, 7, 8, '2026-09-10 12:00:00', 'Cancelled', 3299.00),
  (9, 8, 9, '2026-09-14 19:35:00', 'Completed', 3797.00),
  (10, 2, 3, '2026-09-20 13:15:00', 'Completed', 5497.00),
  (11, 9, 10, '2026-09-27 10:50:00', 'Confirmed', 4398.00),
  (12, 10, 11, '2026-10-03 15:25:00', 'Confirmed', 4149.00),
  (13, 3, 4, '2026-10-09 20:05:00', 'Placed', 3197.00),
  (14, 5, 6, '2026-10-15 08:40:00', 'Placed', 18998.00),
  (15, 11, 12, '2026-10-22 17:55:00', 'Placed', 4997.00);
```

2.7 order_items

```
INSERT INTO order_items (order_item_id, order_id, product_id, quantity, unit_price) VALUES
  (1, 1, 1, 1, 1499.00),
  (2, 1, 8, 1, 599.00),
  (3, 2, 4, 1, 15499.00),
  (4, 2, 19, 1, 1599.00),
  (5, 3, 5, 3, 499.00),
  (6, 3, 6, 1, 1299.00),
  (7, 4, 11, 1, 2299.00),
  (8, 4, 12, 1, 899.00),
  (9, 5, 17, 1, 3299.00),
  (10, 5, 18, 1, 3999.00),
  (11, 6, 2, 1, 1899.00),
  (12, 6, 14, 1, 699.00),
  (13, 7, 9, 1, 1250.00),
  (14, 7, 8, 2, 599.00),
  (15, 8, 15, 1, 3299.00),
  (16, 9, 16, 1, 2799.00),
  (17, 9, 5, 2, 499.00),
  (18, 10, 3, 1, 2499.00),
  (19, 10, 1, 2, 1499.00),
  (20, 11, 13, 1, 3499.00),
  (21, 11, 12, 1, 899.00),
  (22, 12, 10, 1, 850.00),
  (23, 12, 17, 1, 3299.00),
  (24, 13, 7, 1, 1799.00),
  (25, 13, 14, 2, 699.00),
  (26, 14, 4, 1, 14999.00),
  (27, 14, 18, 1, 3999.00),
  (28, 15, 1, 1, 1499.00),
  (29, 15, 2, 1, 1899.00),
  (30, 15, 19, 1, 1599.00);
```

2.8 payments

```
INSERT INTO payments (payment_id, order_id, payment_method, amount, payment_status) VALUES
  (1, 1, 'UPI', 2098.00, 'Paid'),
  (2, 2, 'Card', 17098.00, 'Paid'),
  (3, 3, 'UPI', 2796.00, 'Paid'),
  (4, 4, 'Net Banking', 3198.00, 'Paid'),
  (5, 5, 'Card', 7298.00, 'Paid'),
  (6, 6, 'UPI', 2598.00, 'Paid'),
  (7, 7, 'COD', 2448.00, 'Paid'),
  (8, 8, 'UPI', 3299.00, 'Refunded'),
  (9, 9, 'Card', 3797.00, 'Paid'),
  (10, 10, 'UPI', 5497.00, 'Paid'),
  (11, 11, 'Card', 4398.00, 'Paid'),
  (12, 12, 'UPI', 4149.00, 'Paid'),
  (13, 13, 'COD', 3197.00, 'Pending'),
  (14, 14, 'Card', 18998.00, 'Failed'),
  (15, 14, 'UPI', 18998.00, 'Paid'),
  (16, 15, 'Net Banking', 4997.00, 'Pending');
```

2.9 shipments

```
INSERT INTO shipments (shipment_id, order_id, tracking_no, carrier, delivery_status) VALUES
  (1, 1, 'TRK20260001', 'Delhivery', 'Delivered'),
  (2, 2, 'TRK20260002', 'Blue Dart', 'Delivered'),
  (3, 3, 'TRK20260003', 'Ecom Express', 'Delivered'),
  (4, 4, 'TRK20260004', 'Delhivery', 'Delivered'),
  (5, 5, 'TRK20260005', 'Blue Dart', 'Delivered'),
  (6, 6, 'TRK20260006', 'Delhivery', 'Delivered'),
  (7, 6, 'TRK20260007', 'India Post', 'Delivered'),
  (8, 7, 'TRK20260008', 'Ecom Express', 'Delivered'),
  (9, 9, 'TRK20260009', 'Delhivery', 'Delivered'),
  (10, 10, 'TRK20260010', 'Blue Dart', 'Delivered'),
  (11, 11, 'TRK20260011', 'Ecom Express', 'Shipped'),
  (12, 12, 'TRK20260012', 'Delhivery', 'Packed');
```

2.10 reviews

```
INSERT INTO reviews (review_id, customer_id, product_id, rating, review_text) VALUES
  (1, 1, 1, 5, 'Great sound and battery life.'),
  (2, 1, 8, 4, 'Clear explanations for beginners.'),
  (3, 2, 4, 5, 'Sharp display, good value.'),
  (4, 2, 19, 4, 'Comfortable grip, responsive clicks.'),
  (5, 3, 5, 3, 'Fabric is fine but runs slightly small.'),
  (6, 4, 11, 5, 'Even heating and easy to clean.'),
  (7, 5, 17, 4, 'Solid build quality.'),
  (8, 5, 18, 5, 'Keys feel great for long sessions.'),
  (9, 1, 2, 4, 'Charges fast, a little heavy.'),
  (10, 6, 9, 5, 'Excellent reference for coursework.'),
  (11, 8, 16, 4, 'Light and comfortable.'),
  (12, 2, 1, 2, 'Left earbud stopped working in a month.'),
  (13, 8, 5, 4, 'Good everyday t-shirt.'),
  (14, 2, 3, 3, 'Tracking is accurate, strap is stiff.'),
  (15, 6, 8, 5, 'Practical examples throughout.');
```

Note: Totals. orders.total_amount equals the sum of quantity × unit_price of that order's items, calculated before inserting. Section 4 verifies this with a query.

03 TASK 3

Verifying the Inserted Data

Row counts first, then every table is displayed.

3.1 Row count of every table

```
SELECT 'customers' AS table_name, COUNT(*) AS row_count FROM customers
UNION ALL SELECT 'pincodes', COUNT(*) FROM pincodes
UNION ALL SELECT 'addresses', COUNT(*) FROM addresses
UNION ALL SELECT 'categories', COUNT(*) FROM categories
UNION ALL SELECT 'products', COUNT(*) FROM products
UNION ALL SELECT 'orders', COUNT(*) FROM orders
UNION ALL SELECT 'order_items', COUNT(*) FROM order_items
UNION ALL SELECT 'payments', COUNT(*) FROM payments
UNION ALL SELECT 'shipments', COUNT(*) FROM shipments
UNION ALL SELECT 'reviews', COUNT(*) FROM reviews;
```

OUTPUT

table_name	row_count
customers	12
pincodes	10
addresses	14
categories	6
products	20
orders	15
order_items	30
payments	16
shipments	12
reviews	15

(10 rows, 150 records in total)

3.2 customers

```
SELECT * FROM customers;
```

OUTPUT

customer_id	name	email	phone
1	Aarav Sharma	aarav.sharma@example.com	9876500001
2	Priya Nair	priya.nair@example.com	9876500002
3	Rohan Mehta	rohan.mehta@example.com	9876500003
4	Ananya Iyer	ananya.iyer@example.com	9876500004
5	Karthik Reddy	karthik.reddy@example.com	9876500005
6	Sneha Patil	sneha.patil@example.com	9876500006
7	Vikram Singh	vikram.singh@example.com	9876500007
8	Divya Menon	divya.menon@example.com	9876500008
9	Arjun Das	arjun.das@example.com	9876500009
10	Meera Kulkarni	meera.kulkarni@example.com	9876500010
11	Imran Khan	imran.khan@example.com	9876500011
12	Lakshmi Venkatesh	lakshmi.venkatesh@example.com	NULL

(12 rows)

3.3 pincodes

```
SELECT * FROM pincodes;
```

OUTPUT

pincode	city	state
560001	Bengaluru	Karnataka
560034	Bengaluru	Karnataka
560066	Bengaluru	Karnataka
400001	Mumbai	Maharashtra
110001	New Delhi	Delhi
600001	Chennai	Tamil Nadu
500001	Hyderabad	Telangana
411001	Pune	Maharashtra
700001	Kolkata	West Bengal
380001	Ahmedabad	Gujarat

(10 rows)

3.4 addresses

```
SELECT * FROM addresses;
```

OUTPUT

address_id	customer_id	label	house_street	pincode
1	1	Home	12, 4th Cross, Indiranagar	560001
2	2	Home	45, Residency Road	560034
3	2	Office	7, Tech Park Main Road, Whitefield	560066
4	3	Home	203, Marine Lines Apartments	400001
5	4	Home	88, Gandhi Nagar	600001
6	5	Home	14, Jubilee Hills Road No. 5	500001
7	6	Home	31, Model Colony	411001
8	7	Home	9, Connaught Circus	110001
9	8	Home	56, Salt Lake Sector 2	700001
10	9	Home	22, Navrangpura Society	380001
11	10	Home	5, Koregaon Park Lane	411001
12	11	Home	101, Park Street	700001
13	12	Home	18, Brigade Road	560001
14	1	Office	3, MG Road Business Centre	560001

(14 rows)

3.5 categories

```
SELECT * FROM categories;
```

OUTPUT

category_id	category_name	description
1	Electronics	Phones, audio devices and accessories
2	Clothing	Men's and women's apparel
3	Books	Academic and general reading
4	Home & Kitchen	Appliances and cookware
5	Sports & Fitness	Exercise equipment and activewear
6	Gaming	Controllers, keyboards and gaming accessories

(6 rows)

3.6 products

```
SELECT * FROM products;
```

OUTPUT

product_id	category_id	name	price	stock_qty
1	1	Wireless Bluetooth Earbuds	1499.00	120
2	1	20000 mAh Power Bank	1899.00	85
3	1	Smart Fitness Band	2499.00	60
4	1	27-inch IPS Monitor	14999.00	25
5	2	Cotton Round-Neck T-Shirt	499.00	300
6	2	Slim Fit Denim Jeans	1299.00	150
7	2	Hooded Sweatshirt	1799.00	90
8	3	SQL Fundamentals for Beginners	599.00	200
9	3	Database System Concepts	1250.00	75
10	3	Introduction to Game Design	850.00	0
11	4	Non-Stick Cookware Set	2299.00	55
12	4	Electric Kettle 1.5 L	899.00	110
13	4	Mixer Grinder 750 W	3499.00	35
14	5	Yoga Mat 6 mm	699.00	140
15	5	Adjustable Dumbbell Pair	3299.00	30
16	5	Running Shoes	2799.00	70
17	6	Wireless Game Controller	3299.00	45
18	6	Mechanical Gaming Keyboard	3999.00	38
19	6	Gaming Mouse	1599.00	80
20	6	Large Mouse Pad	399.00	60

(20 rows)

3.7 orders

```
SELECT * FROM orders;
```

OUTPUT

order_id	customer_id	address_id	order_date	status	total_amount
1	1	1	2026-08-03 10:15:00	Completed	2098.00
2	2	2	2026-08-07 14:40:00	Completed	17098.00
3	3	4	2026-08-12 09:05:00	Completed	2796.00
4	4	5	2026-08-19 18:22:00	Completed	3198.00
5	5	6	2026-08-25 21:10:00	Completed	7298.00
6	1	14	2026-09-02 11:30:00	Completed	2598.00
7	6	7	2026-09-06 16:45:00	Completed	2448.00
8	7	8	2026-09-10 12:00:00	Cancelled	3299.00
9	8	9	2026-09-14 19:35:00	Completed	3797.00
10	2	3	2026-09-20 13:15:00	Completed	5497.00
11	9	10	2026-09-27 10:50:00	Confirmed	4398.00
12	10	11	2026-10-03 15:25:00	Confirmed	4149.00
13	3	4	2026-10-09 20:05:00	Placed	3197.00
14	5	6	2026-10-15 08:40:00	Placed	18998.00
15	11	12	2026-10-22 17:55:00	Placed	4997.00

(15 rows)

3.8 order_items

```
SELECT * FROM order_items;
```

OUTPUT

order_item_id	order_id	product_id	quantity	unit_price
1	1	1	1	1499.00
2	1	8	1	599.00
3	2	4	1	15499.00
4	2	19	1	1599.00
5	3	5	3	499.00
6	3	6	1	1299.00
7	4	11	1	2299.00
8	4	12	1	899.00
9	5	17	1	3299.00
10	5	18	1	3999.00
11	6	2	1	1899.00
12	6	14	1	699.00
13	7	9	1	1250.00
14	7	8	2	599.00
15	8	15	1	3299.00
16	9	16	1	2799.00
17	9	5	2	499.00
18	10	3	1	2499.00
19	10	1	2	1499.00
20	11	13	1	3499.00
21	11	12	1	899.00
22	12	10	1	850.00
23	12	17	1	3299.00
24	13	7	1	1799.00
25	13	14	2	699.00
26	14	4	1	14999.00
27	14	18	1	3999.00
28	15	1	1	1499.00
29	15	2	1	1899.00
30	15	19	1	1599.00

(30 rows)

3.9 payments

```
SELECT * FROM payments;
```

OUTPUT

payment_id	order_id	payment_method	amount	payment_status
1	1	UPI	2098.00	Paid
2	2	Card	17098.00	Paid
3	3	UPI	2796.00	Paid
4	4	Net Banking	3198.00	Paid
5	5	Card	7298.00	Paid
6	6	UPI	2598.00	Paid
7	7	COD	2448.00	Paid
8	8	UPI	3299.00	Refunded
9	9	Card	3797.00	Paid
10	10	UPI	5497.00	Paid
11	11	Card	4398.00	Paid
12	12	UPI	4149.00	Paid
13	13	COD	3197.00	Pending
14	14	Card	18998.00	Failed
15	14	UPI	18998.00	Paid
16	15	Net Banking	4997.00	Pending

(16 rows)

3.10 shipments

```
SELECT * FROM shipments;
```

OUTPUT

shipment_id	order_id	tracking_no	carrier	delivery_status
1	1	TRK20260001	Delhivery	Delivered
2	2	TRK20260002	Blue Dart	Delivered
3	3	TRK20260003	Ecom Express	Delivered
4	4	TRK20260004	Delhivery	Delivered
5	5	TRK20260005	Blue Dart	Delivered
6	6	TRK20260006	Delhivery	Delivered
7	6	TRK20260007	India Post	Delivered
8	7	TRK20260008	Ecom Express	Delivered
9	9	TRK20260009	Delhivery	Delivered
10	10	TRK20260010	Blue Dart	Delivered
11	11	TRK20260011	Ecom Express	Shipped
12	12	TRK20260012	Delhivery	Packed

(12 rows)

3.11 reviews

```
SELECT * FROM reviews;
```

OUTPUT

review_id	customer_id	product_id	rating	review_text
1	1	1	5	Great sound and battery life.
2	1	8	4	Clear explanations for beginners.
3	2	4	5	Sharp display, good value.
4	2	19	4	Comfortable grip, responsive clicks.
5	3	5	3	Fabric is fine but runs slightly small.
6	4	11	5	Even heating and easy to clean.
7	5	17	4	Solid build quality.
8	5	18	5	Keys feel great for long sessions.
9	1	2	4	Charges fast, a little heavy.
10	6	9	5	Excellent reference for coursework.
11	8	16	4	Light and comfortable.
12	2	1	2	Left earbud stopped working in a month.
13	8	5	4	Good everyday t-shirt.
14	2	3	3	Tracking is accurate, strap is stiff.
15	6	8	5	Practical examples throughout.

(15 rows)

04 TASK 4

Checking Constraints and Relationships on the Data

Each check counts rule violations. A correct data set returns 0 for every line.

```
SELECT 'orders with no customer' AS check_name, COUNT(*) AS violations
FROM orders o LEFT JOIN customers c ON c.customer_id = o.customer_id
WHERE c.customer_id IS NULL
UNION ALL
SELECT 'orders with no delivery address', COUNT(*)
FROM orders o LEFT JOIN addresses a ON a.address_id = o.address_id
WHERE a.address_id IS NULL
UNION ALL
SELECT 'orders using another customer''s address', COUNT(*)
FROM orders o JOIN addresses a ON a.address_id = o.address_id
WHERE a.customer_id <> o.customer_id
UNION ALL
SELECT 'order lines with no product', COUNT(*)
FROM order_items i LEFT JOIN products p ON p.product_id = i.product_id
WHERE p.product_id IS NULL
UNION ALL
SELECT 'payments with no order', COUNT(*)
FROM payments y LEFT JOIN orders o ON o.order_id = y.order_id
WHERE o.order_id IS NULL
UNION ALL
SELECT 'shipments with no order', COUNT(*)
FROM shipments s LEFT JOIN orders o ON o.order_id = s.order_id
WHERE o.order_id IS NULL
UNION ALL
SELECT 'reviews with no customer or product', COUNT(*)
FROM reviews r
WHERE r.customer_id NOT IN (SELECT customer_id FROM customers)
   OR r.product_id  NOT IN (SELECT product_id  FROM products)
UNION ALL
SELECT 'orders whose total differs from their lines', COUNT(*)
FROM orders o
WHERE o.total_amount <> (SELECT SUM(i.quantity * i.unit_price)
                         FROM order_items i WHERE i.order_id = o.order_id);
```

OUTPUT

check_name	violations
orders with no customer	0
orders with no delivery address	0
orders using another customer's address	0
order lines with no product	0
payments with no order	0
shipments with no order	0
reviews with no customer or product	0
orders whose total differs from their lines	0

(8 rows)

Explanation: The first seven checks look for orphan rows (a child row whose parent does not exist), including the rule that an order must be delivered to an address belonging to the same customer. The last check confirms that the stored order total matches its line items. All eight return 0, so the data respects every relationship.

> **Note: Conclusion.** 150 records across ten tables now simulate a working shop. The same data is used for every query in Assignments 8 and 9.